
Celler

Complete Documentation

Cryptocurrency Exchange Platform
Nigerian Market Focus

Version 0.1.0 | Frontend Prototype | April 3, 2026

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Celler — Full Product Overview

Version: 0.1.0 (Prototype)

Date: April 3, 2026

Status: Frontend Prototype — No Backend

Project Type: Crypto Exchange (CEX) — Nigerian Market Focus

1. Product Summary

Celler is a cryptocurrency exchange platform targeting the Nigerian market. It enables users to buy, sell, swap, deposit, and withdraw cryptocurrencies and Nigerian Naira (NGN). The platform also includes a gift card trading feature, a referral rewards program, and KYC verification tiers.

An admin dashboard provides oversight of users, transactions, gift card orders, KYC approvals, referral management, and system-wide settings.

2. Classification

Attribute	Value
Backend	None (all data in localStora
Auth	Mock (localStorage-backed, n
APIs	None (zero real API calls)
Database	None (localStorage as pseudo
Framework	Vite 5 + React 18 + TypeScri
Routing	React Router 6 (client-side)
State	React Context + localStorage

3. Core Use Cases

#	Use Case	Description
2	Crypto Swap	Exchange one cryptocurrency
3	Withdrawal	Send crypto to external addr
4	Receive	Display QR codes for crypto
5	Transaction History	View, filter, and search all
6	Gift Card Trading	Sell gift cards (Amazon, iTu
7	Referral Program	Earn NGN rewards for referri
8	KYC Verification	3-tier identity verification
9	Profile & Security	Manage profile, password, PI

#	Use Case	Description
10	Admin Oversight	Monitor users, transactions,

4. User Types

Role	Description	Access Level
Regular User	Verified or unverified accou	Full user app (dashboard, sw
Admin (Super Admin)	Platform operator	Full admin dashboard (users,

5. System Architecture

Actual Architecture (Current)



Intended Architecture (Production)



6. Supported Assets

Asset	Type	Default Balance	Price (USD)
ETH	Crypto	0.15	\$2,620.51
USDT	Stablecoin	250	\$1.00
BNB	Crypto	1.2	\$640.14
SOL	Crypto	5	\$192.00
TRX	Crypto	1,000	\$0.30
USDC	Stablecoin	100	\$1.00
NGN	Fiat	50,000	\$0.00066

7. Key Constants & Configuration

Parameter	Value	Location
Default Trade Fee	1.5%	data/feeConfig.ts
Default Swap Fee	0.8% (config) / 0.2% (hardco	data/feeConfig.ts + `pages
Default Withdrawal Fee	0.5%	data/feeConfig.ts
NGN Withdrawal Fee	100 NGN (flat)	pages/Withdraw.tsx
Crypto Withdrawal Fee	\$0.50 (flat)	pages/Withdraw.tsx
Referral Reward	5,000 NGN	lib/referral.ts
Weekly Referral Withdrawal L	10,000 NGN	lib/referral.ts
Transaction PIN	1234 (hardcoded)	pages/Withdraw.tsx
2FA Mock Secret	JBSWY3DPEHPK3PPX	pages/TwoFactor.tsx

8. Module Status Summary

Module	Status	Notes
Wallet/Dashboard	Partial	UI complete, no real price f
Swap	Partial	UI + logic complete, no real
Withdrawal	Partial	Multi-step flow complete, no
Receive	Partial	Static QR codes, no real dep
Transaction History	Partial	Mock data only
Gift Card Trading	Partial	Full UI flow, no real proces
Referral Program	Partial	UI + mock store, no real tra
KYC Verification	Partial	UI complete, no real verific
Profile & Security	Partial	UI complete, no real backend
Admin Dashboard	Partial	Mock data only, no real cont
Notifications	Mock	Hardcoded notifications
Support	Partial	Static page, no real chat

9. Critical Gaps

Gap	Severity	Description
No Real Auth	Critical	Anyone can access any route
No Admin Auth	Critical	Admin routes have no guards
Buy/Sell Missing	High	Routes not defined, links br
Hardcoded PIN	High	1234 is hardcoded in code
No Real Price Feeds	High	All prices are static
No Real Blockchain	High	No wallet addresses, no tran
Unused Dependencies	Medium	TanStack Query, next-themes
TODO in HTML	Low	Title still says "Lovable Ap

Celler — Module Breakdown (Epics)

EPIC 1: Authentication & Onboarding

Description: User registration, login, onboarding flow, and session management.

Key Features:

- Splash screen with auto-redirect
- Welcome carousel (3 slides)
- Sign In (email + password)
- Sign Up (4-step: email → OTP → password → success)
- Session persistence via localStorage

Dependencies: None (self-contained)

Status: Mocked

Aspect	Detail
Password Reset	Not implemented
Email Verification	Mock — any 6-digit OTP accep
Session Security	None — no token expiry, no r
Auto-login	Yes — first visit creates mo

EPIC 2: Wallet & Dashboard

Description: Main user-facing screen showing portfolio balance, quick actions, and asset list.

Key Features:

- Total portfolio balance (USD + NGN)
- Quick action buttons (Buy, Sell, Send, Receive, Swap, More)
- Asset list with balances and 24h change
- Promo banner carousel
- Manage visible assets
- Individual coin detail pages

Dependencies:

- lib/crypto.ts (wallet store, coin data)
- AuthContext (user identity)

Status: Partial

Aspect	Detail
Buy Flow	Missing — route not defined
Sell Flow	Missing — route not defined
Real-time Updates	None

Aspect	Detail
Portfolio Chart	None

EPIC 3: Crypto Swap

Description: Exchange one cryptocurrency for another with rate calculation and fee display.

Key Features:

- Coin pair selection (from/to)
- Real-time rate calculation
- Fee display (0.2% hardcoded)
- Slippage tolerance display (0.5%)
- Review screen before confirmation
- Success animation with transaction timeline

Dependencies:

- lib/crypto.ts (coin prices, wallet balances)
- AuthContext

Status: Partial

Aspect	Detail
Rate Source	Hardcoded prices
Execution	Mock — updates localStorage
Slippage Handling	Display only, no real enforc

EPIC 4: Withdrawal

Description: Multi-step withdrawal flow for crypto and NGN with authentication.

Key Features:

- Amount entry with custom numpad
- Recipient selection (crypto address or bank account)
- Review summary with fee breakdown
- Authentication step (FaceID mock or PIN)
- Success animation with timeline
- Beneficiary management

Dependencies:

- lib/crypto.ts (wallet, banks, beneficiaries)
- AuthContext

Status: Partial

Aspect	Detail
Bank Transfer Integration	None
PIN Security	Hardcoded 1234

Aspect	Detail
FacelD	Mock — 2-second timer
Address Validation	None
Withdrawal Limits	Not enforced

EPIC 5: Receive & Deposit

Description: Display deposit addresses (QR codes) for crypto and bank details for NGN.

Key Features:

- QR code display for crypto addresses
- Copy-to-clipboard for addresses
- NGN bank details display
- Network selection for multi-network coins

Dependencies:

- lib/crypto.ts (mock addresses)

Status: Partial

Aspect	Detail
Deposit Detection	None
QR Code Generation	Static images
Network Support	Display only

EPIC 6: Transaction History

Description: View, filter, and search all user transactions.

Key Features:

- Transaction list with type icons
- Filter by type (buy, sell, swap, deposit, withdraw, receive, send)
- Filter by status (completed, pending, failed)
- Search functionality
- Transaction detail view
- Receipt generation (printable/shareable)

Dependencies:

- lib/crypto.ts (transaction store)

Status: Partial

Aspect	Detail
Pagination	None
Export	Receipt view only
Real-time Updates	None

EPIC 7: Gift Card Trading

Description: Sell gift cards for NGN payout with brand selection and rate tiers.

Key Features:

- Gift card marketplace (8 brands across categories)
- Sell flow: amount → upload card image → review → success
- Tiered rate structure (55-75% of face value)
- Order history
- Admin approval workflow

Dependencies:

- data/giftcardData.ts (brands, rates, order store)
- lib/crypto.ts (NGN wallet for payout)

Status: Partial

Aspect	Detail
Rate Engine	Client-side calculation
Payout	Mock — credited to localStor
Fraud Detection	None
Admin Processing	Mock status updates

EPIC 8: Referral Program

Description: Earn NGN rewards for referring new users with withdrawal management.

Key Features:

- Referral code and shareable link
- Referral tracking (signed up, verified, traded)
- Earnings dashboard
- Withdrawal requests with weekly limit
- Referral history

Dependencies:

- lib/referral.ts (referral store, config)

Status: Partial

Aspect	Detail
Link Generation	Static URL
Withdrawal Processing	Admin approval required (moc
Weekly Limit	Enforced in client-side stor

EPIC 9: KYC Verification

Description: 3-tier identity verification system with increasing transaction limits.

Key Features:

- Tier 1: BVN + Liveness Check (auto-approval)
- Tier 2: Government ID + Address Proof (auto-approval)
- Tier 3: Employment + Risk Assessment (manual approval)
- Tier limits display
- Verification status tracking

Dependencies:

- data/kycConfig.ts (tier limits)

Status: Partial

Aspect	Detail
Liveness Check	Not implemented
BVN Verification	Not implemented
Admin Review	Mock workflow
Limit Enforcement	Display only

EPIC 10: Profile & Security

Description: User profile management, security settings, and app preferences.

Key Features:

- Profile editing (name, username, phone)
- Password change
- Transaction PIN update
- 2FA setup
- Bank account management
- Appearance/theme settings
- Support & help

Dependencies:

- AuthContext
- lib/crypto.ts (user data, banks)

Status: Partial

Aspect	Detail
PIN Update	3-step flow, mock validation
2FA Setup	Mock QR code and secret
Bank Management	localStorage only
Theme	Working (dark/light/system)

EPIC 11: Notifications

Description: In-app notification system for users and admin push notification composer.

Key Features:

- User notification list (4 hardcoded)

- Admin push notification composer
- Notification history (admin view)

Dependencies: None (fully mocked)

Status: Mocked

Aspect	Detail
Real-time	None
Push Integration	None
Read/Unread	Not tracked

EPIC 12: Admin Dashboard

Description: Platform management interface for monitoring users, transactions, and system settings.

Key Features:

- Overview dashboard with stats and revenue chart
- Customer management (list, detail, KYC status)
- Transaction/order monitoring
- Gift card order approval/rejection
- Referral management and configuration
- KYC approval queue
- Push notification composer
- System settings (fees, KYC limits, referral config)

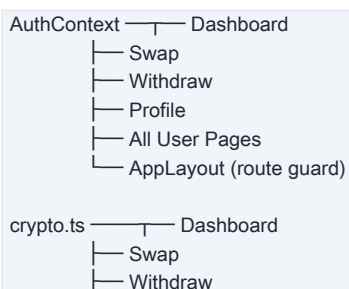
Dependencies:

- data/adminMockData.ts (all mock data)
- All data stores (read access)

Status: Partial

Aspect	Detail
Real-time Data	None
Action Execution	Mock — no real effects
Audit Trail	None
Role Management	Single hardcoded admin
Fraud Detection	None

Module Dependency Map



	└─ Receive
	└─ History
	└─ CoinDetail
	└─ TransactionDetail
	└─ Receipt
	└─ BankDetails
	└─ GiftcardSell (NGN payout)
referral.ts	└─ Referral (user)
	└─ AdminReferrals
giftcardData	└─ Giftcards
	└─ GiftcardSe
... [truncated]	

PRD — Authentication & Onboarding

1. Overview

The Authentication & Onboarding module handles user registration, login, session management, and the initial app experience. It is the entry point for all users.

2. Goals & Objectives

Goal	Metric
Secure user accounts	Password validation, 2FA sup
Clear onboarding	Users understand value propo

3. User Personas

Persona	Description
Returning User	Existing user, needs quick a
Demo User	Auto-created mock user for p

4. User Stories

- US-1.1: View Splash Screen
- US-1.2: View Welcome Carousel
- US-1.3: Sign In to Existing Account
- US-1.4: Create New Account
- US-1.5: Stay Logged In Across Sessions
- US-1.6: Log Out

5. User Flows

Sign-In Flow

Splash (2.5s) → Check isLoggedIn → If yes: Dashboard
→ If no: Welcome → Sign In → Dashboard

Sign-Up Flow

Welcome → Sign Up → Step 1: Email → Step 2: OTP (any 6-digit)
 → Step 3: Password (8+ chars, uppercase, special char)
 → Step 4: Success → Sign In or Dashboard

Logout Flow

Profile → Log Out → Clear localStorage → Redirect to Sign In

6. Functional Requirements

ID	Requirement	Priority
FR-1.2	Welcome carousel has 3 navig	High
FR-1.3	Sign In validates email and	High
FR-1.4	Sign Up is a 4-step flow wit	High
FR-1.5	OTP step accepts any 6-digit	Medium
FR-1.6	Password requires 8+ chars,	High
FR-1.7	Session persists via localSt	High
FR-1.8	Auto-login creates mock user	Medium
FR-1.9	Logout clears session and re	High
FR-1.10	Authenticated routes redirec	High

7. Non-Functional Requirements

ID	Requirement
NFR-1.2	Form inputs are accessible (
NFR-1.3	Password field masks input
NFR-1.4	Session data is stored secur

8. Edge Cases / Unhappy Paths

Scenario	Expected Behavior
Wrong password on sign-in	Show "Invalid credentials" e
Empty fields on sign-in	Show validation error
Password mismatch on sign-up	Show "Passwords do not match
Weak password on sign-up	Show password requirements n
OTP with fewer than 6 digits	Disable "Continue" button
Refresh during sign-up flow	Lose progress (no draft save

9. Acceptance Criteria

Given I open the app for the first time
When the splash screen displays
Then I see the Celler branding
And I am redirected to the welcome page after 2.5 seconds

Given I am on the sign-in page
When I enter valid credentials
Then I am redirected to the dashboard
And my session is persisted

Given I am logged in
When I close and reopen the app
Then I am automatically on the dashboard

10. API / Data Requirements

Item	Status	Details
User Authentication	Missing Backend	Currently localStorage only
Session Management	Missing Backend	No token, no expiry
Email OTP	Mocked	Any 6-digit code accepted
Password Reset	Missing	Not implemented
Auto-login	Mocked	Creates MOCK_USER on first

Data Model (Intended)

```
interface User {  
  id: string;  
  email: string;  
  passwordHash: string;  
  fullName: string;  
  username: string;  
  phone: string;  
  createdAt: Date;  
  lastLoginAt: Date;  
  status: 'active' | 'suspended' | 'deleted';  
}  
  
interface Session {  
  id: string;  
  userId: string;  
  token: string;  
  expiresAt: Date;  
  deviceInfo: string;  
}
```

11. UI/UX Notes

- Splash: Full-screen gradient with centered logo

- Welcome: Carousel with pagination dots, skip option implied
- Sign In: Clean form, "Forgot Password?" link (non-functional)
- Sign Up: Step indicator at top, progress visible
- Error states: Red text below input fields
- Loading states: Button shows spinner during submission

12. Metrics / Success Criteria

Metric	Target
Sign-in success rate (first	90%
Time from splash to dashboar	< 30s (new user)
Session persistence reliabil	100%

INTENDED BEHAVIOR vs CURRENT IMPLEMENTATION

Aspect	Intended	Current
Email Verification	Real OTP via email/SMS	Any 6-digit code accepted
Password Storage	Hashed (bcrypt)	Plaintext in localStorage
Session Security	JWT with expiry, refresh tok	Boolean flag in localStorage
Forgot Password	Email-based reset flow	Not implemented
Auto-login	Remember device option	Always auto-logs in with moc
Rate Limiting	Brute-force protection	None

PRD — Wallet & Dashboard

1. Overview

The Wallet & Dashboard module is the primary user-facing screen. It displays the user's portfolio balance, asset holdings, quick action shortcuts, and promotional content.

2. Goals & Objectives

Goal	Metric
Enable quick access to core	All key actions reachable in
Display accurate asset value	Real-time price feeds (inten

3. User Personas

Persona	Description
Casual User	Occasionally checks portfolio
New User	Exploring the dashboard, lea

4. User Stories

- US-2.1: View Portfolio Balance
- US-2.2: View Asset List
- US-2.3: Access Quick Actions
- US-2.4: Manage Visible Assets
- US-2.5: View Coin Details

5. User Flows

Dashboard Load Flow

AppLayout (auth guard) → Dashboard → Fetch wallet → Calculate total → Render

Quick Action Flow

Dashboard → Tap action button → Navigate to feature page

Manage Assets Flow

Dashboard → Tap "Manage Assets" → Toggle coins → Save → Return to Dashboard

Coin Detail Flow

Dashboard → Tap coin → CoinDetail → View balance, actions, history

6. Functional Requirements

ID	Requirement	Priority
FR-2.2	Display individual coin bala	High
FR-2.3	Show 24h price change percen	Medium
FR-2.4	Quick action buttons: Buy, S	High
FR-2.5	Promo banner carousel with a	Low
FR-2.6	Toggle coin visibility on da	Medium
FR-2.7	Navigate to coin detail page	High
FR-2.8	Mobile bottom navigation + d	High

7. Non-Functional Requirements

ID	Requirement
NFR-2.2	Balance calculations are acc
NFR-2.3	UI is responsive across mobi
NFR-2.4	Coin icons load from externa

8. Edge Cases / Unhappy Paths

Scenario	Expected Behavior
User has zero balance for a	Show 0.00 with appropriate f
External price feed unavaila	Show last known price with "
User hides all coins	Show empty state with "Add a
Network disconnect	Show cached data with offlin

9. Acceptance Criteria

Given I am logged in
 When I navigate to the dashboard
 Then I see my total balance in USD and NGN
 And I see a list of my coins with balances

And I see quick action buttons
And tapping a coin navigates to its detail page

10. API / Data Requirements

Item	Status	Details
Coin Prices	Mocked	Hardcoded in lib/crypto.ts
24h Price Change	Mocked	Hardcoded percentages
Portfolio Total	Computed	Client-side calculation
Visible Coins	Mocked	localStorage store
Promo Banners	Hardcoded	3 static banners

Data Model (Intended)

```
interface Wallet {
  userId: string;
  balances: { [coinId: string]: number };
  totalUsd: number;
  totalNgn: number;
  lastUpdated: Date;
}

interface CoinPrice {
  coinId: string;
  priceUsd: number;
  change24h: number;
  lastUpdated: Date;
  source: string;
}
```

11. UI/UX Notes

- Balance: Large, prominent display at top
- Quick actions: Icon grid with labels
- Asset list: Scrollable, each row shows icon, name, balance, value, change
- Promo banners: Horizontal carousel with auto-advance
- Mobile: Bottom tab nav (Home, Swap, History, Profile)
- Desktop: Left sidebar + top bar

12. Metrics / Success Criteria

Metric	Target
Balance accuracy	100%
Quick action tap-through rat	40%
User engagement (daily activ	60% of registered users

INTENDED BEHAVIOR vs CURRENT IMPLEMENTATION

Aspect	Intended	Current
Balance Updates	Real-time from blockchain/wa	localStorage only
Buy Flow	Integrated payment gateway	Missing — route not defined
Sell Flow	Integrated off-ramp	Missing — route not defined
Portfolio Chart	Historical balance chart	Not implemented
Bill Payments	Feature in development	Waitlist banner only

PRD — Crypto Swap

1. Overview

The Swap module enables users to exchange one cryptocurrency for another with transparent rate calculation, fee display, and confirmation flow.

2. Goals & Objectives

- Enable seamless crypto-to-crypto exchange
- Display transparent pricing with fees and slippage
- Provide confirmation before execution

3. User Personas

Persona	Description
Arbitrage User	Exploits rate differences be

4. User Stories

- US-3.1: Swap Cryptocurrency
- US-3.2: View Swap Rate and Fees
- US-3.3: Swap Coin Selection

5. User Flows

Dashboard → Swap → Select "from" coin → Enter amount
 → Select "to" coin → View rate & fee → Review → Confirm
 → Success animation → Wallet updated → History updated

6. Functional Requirements

ID	Requirement	Priority
FR-3.2	Calculate exchange rate in r	High
FR-3.3	Display fee (0.2% hardcoded)	High
FR-3.4	Display slippage tolerance (	Medium

ID	Requirement	Priority
FR-3.5	Review screen before confirm	High
FR-3.6	Success animation with trans	Medium
FR-3.7	Update wallet balances post-	High
FR-3.8	Record transaction in histor	High

7. Non-Functional Requirements

- Rate calculation completes in < 100ms
- UI remains responsive during calculation
- Balances update atomically (no partial updates)

8. Edge Cases

Scenario	Expected Behavior
Same coin selected for both	Prevent selection
Amount exceeds balance	Show "Insufficient balance"
Zero or negative amount	Disable confirm button

9. API / Data Requirements

Item	Status	Details
Swap Execution	Mocked	Client-side localStorage upd
Fee Calculation	Partial	0.2% hardcoded (config says
Rate Source	Mocked	Static ratio of hardcoded pr

10. UI/UX Notes

- Large input field for amount
- Coin selector with search
- Swap button between from/to selectors
- Review modal with full breakdown
- Animated success with timeline steps

11. Metrics

Metric	Target
Average swap time	< 30 seconds

Metric	Target
Error rate (insufficient bal	< 15%

INTENDED vs CURRENT

Aspect	Intended	Current
Fee	Configurable (0.8%)	Hardcoded 0.2%
Execution	Real blockchain/DEX	localStorage update
Slippage	Real enforcement	Display only

PRD — Withdrawal

1. Overview

Multi-step withdrawal flow for cryptocurrency and NGN with authentication (FaceID/PIN), fee calculation, and beneficiary management.

2. Goals & Objectives

- Secure withdrawals with multi-step authentication
- Transparent fee breakdown
- Support both crypto and fiat withdrawals

3. User Personas

Persona	Description
Fiat Withdrawer	Withdraws NGN to bank account

4. User Stories

- US-4.1: Withdraw Cryptocurrency
- US-4.2: Withdraw NGN to Bank
- US-4.3: Add Bank Account
- US-4.4: Authenticate Withdrawal

5. User Flows

CoinDetail/Withdraw → Step 1: Enter amount (custom numpad)
→ Step 2: Enter recipient (address or bank)
→ Step 3: Review summary + fees
→ Step 4: Authenticate (FaceID 2s mock or PIN)
→ Step 5: Success animation + timeline
→ Wallet updated + History updated

6. Functional Requirements

ID	Requirement	Priority
FR-4.2	Crypto address input with va	High
FR-4.3	Bank account selection for N	High
FR-4.4	Fee display (NGN: 100 flat,	High
FR-4.5	Review summary before confir	High
FR-4.6	FacelD authentication (mock	Medium
FR-4.7	PIN authentication (4-digit)	High
FR-4.8	Success animation with timel	Medium
FR-4.9	Add/remove bank accounts	Medium
FR-4.10	Beneficiary management	Low

7. Non-Functional Requirements

- Withdrawal processing completes in < 3s (mock)
- PIN input masks digits
- Address validation for crypto formats
- Atomic balance updates

8. Edge Cases

Scenario	Expected Behavior
Invalid crypto address	Show format validation error
Wrong PIN	Show error, allow retry
Network failure during auth	Retry or cancel gracefully
Withdrawal amount = 0	Disable continue

9. API / Data Requirements

Item	Status	Details
Crypto Transfer	Missing Backend	No blockchain integration
Bank Transfer	Missing Backend	No payment processor
Address Validation	Missing	No format checking
Transaction PIN	Hardcoded	1234 in code
FacelD	Mocked	2-second timer

10. UI/UX Notes

- Custom numpad instead of keyboard for amount
- Clear fee breakdown on review
- Authentication step with two options
- Animated success with step-by-step timeline

11. Metrics

Metric	Target
Authentication failure rate	< 5%
Average withdrawal time	< 60 seconds

INTENDED vs CURRENT

Aspect	Intended	Current
Bank Transfer	Real payment gateway	Mock deduction
Address Validation	Format + checksum validation	None
PIN Security	Hashed, user-set	Hardcoded 1234
FaceID	Real biometric API	2-second timer
Withdrawal Limits	Enforced by KYC tier	Not enforced

PRD — Transaction History

1. Overview

Complete transaction history with filtering, searching, detail views, and receipt generation.

2. Goals & Objectives

- Provide full transparency of all user transactions
- Enable easy filtering and searching
- Generate shareable receipts

3. User Personas

Persona	Description
Dispute Resolver	Needs receipts for support t

4. User Stories

- US-6.1: View Transaction History
- US-6.2: Filter Transactions
- US-6.3: Search Transactions
- US-6.4: View Transaction Details
- US-6.5: View and Share Receipt

5. User Flows

Dashboard → History → View list → Filter/Search → Tap transaction
→ Detail view → View receipt → Share/Print

6. Functional Requirements

ID	Requirement	Priority
FR-6.2	Filter by type (buy, sell, s	High
FR-6.3	Filter by status (completed,	Medium

ID	Requirement	Priority
FR-6.4	Search by keyword	Medium
FR-6.5	Transaction detail view with	High
FR-6.6	Receipt generation with zigz	Medium
FR-6.7	Share/print receipt	Low

7. Non-Functional Requirements

- List renders smoothly with 100+ transactions
- Filter/search responds in < 200ms
- Receipt is printable and shareable

8. Edge Cases

Scenario	Expected Behavior
Search yields no results	Show "No matching transactio
Transaction has missing data	Graceful fallback display
Very long addresses	Truncate with copy option

9. API / Data Requirements

Item	Status	Details
Transaction Detail	Mocked	Derived from mock store
Receipt Generation	Implemented	Client-side rendering
Pagination	Missing	No pagination implemented
Export (CSV/PDF)	Missing	Not implemented

10. UI/UX Notes

- Transaction type icons with color coding
- Status badges (completed = green, pending = yellow, failed = red)
- Zigzag receipt card design
- Mobile-optimized list with pull-to-refresh (intended)

11. Metrics

Metric	Target
Filter/search accuracy	100%

Metric	Target
Receipt generation success	100%

PRD — Gift Card Trading

1. Overview

Gift card marketplace allowing users to sell gift cards (Amazon, iTunes, Steam, etc.) for NGN payout with tiered rates and admin approval workflow.

2. Goals & Objectives

- Enable gift card to cash conversion
- Transparent tiered rate structure
- Admin-controlled approval for fraud prevention

3. User Personas

Persona	Description
Repeat Seller	Regularly sells gift cards

4. User Stories

- US-7.1: Browse Gift Card Brands
- US-7.2: Sell a Gift Card
- US-7.3: View Gift Card Orders

5. User Flows

Dashboard → Giftcards → Browse brands → Select brand
 → Enter amount → Select tier → Upload card image
 → Review order → Submit → Pending status
 → Admin approves → NGN credited to wallet

6. Functional Requirements

ID	Requirement	Priority
FR-7.2	Amount entry with tier selec	High
FR-7.3	NGN payout calculation (amou	High

ID	Requirement	Priority
FR-7.4	Card image upload	High
FR-7.5	Order review before submissi	High
FR-7.6	Order history with status tr	High
FR-7.7	Admin approval/rejection wor	High
FR-7.8	NGN payout on approval	High

7. Non-Functional Requirements

- Image upload supports JPG/PNG up to 5MB
- Rate calculation is accurate to 2 decimal places
- Order status updates in real-time (intended)

8. Edge Cases

Scenario	Expected Behavior
Duplicate card submission	Admin detects and rejects
Image upload fails	Show error, allow retry
Amount below minimum	Show minimum amount error

9. API / Data Requirements

Item	Status	Details
Rate Tiers	Mocked	55-75% of face value
Image Upload	Mocked	No real upload
Order Processing	Mocked	localStorage store
NGN Payout	Mocked	Credited to localStorage wal
Fraud Detection	Missing	No duplicate detection

10. UI/UX Notes

- Brand grid with logos from worldvectorlogo.com
- Clear rate display per tier
- Multi-step sell flow with progress indicator
- Status badges on order history

11. Metrics

Metric	Target
Admin approval time	< 24 hours
Fraud rejection rate	< 5%

PRD — Referral Program

1. Overview

Users earn NGN rewards for referring new users. Includes referral tracking, earnings dashboard, and withdrawal management with weekly limits.

2. Goals & Objectives

- Drive user acquisition through referrals
- Transparent earnings tracking
- Controlled payout via admin approval

3. User Stories

- US-8.1: View Referral Code and Link
- US-8.2: Track Referral Earnings
- US-8.3: Withdraw Referral Earnings

4. Functional Requirements

ID	Requirement	Priority
FR-8.2	Display shareable referral I	High
FR-8.3	Copy code/link to clipboard	High
FR-8.4	Show total earned, withdrawn	High
FR-8.5	Show referral list with stat	High
FR-8.6	Withdrawal request with week	High
FR-8.7	Admin approval for withdrawa	High

5. API / Data Requirements

Item	Status	Details
Earnings Calculation	Mocked	5,000 NGN per referral
Withdrawal Requests	Mocked	localStorage with admin appr
Weekly Limit Enforcement	Implemented	Client-side check
Real Referral Attribution	Missing Backend	No tracking system

6. Metrics

Metric	Target
Weekly withdrawal utilizatio	< 80% of limit



PRD — KYC Verification

1. Overview

3-tier identity verification system with increasing transaction limits. Supports BVN, government ID, and employment verification.

2. User Stories

- US-9.1: View KYC Tiers and Limits
- US-9.2: Complete KYC Verification

3. Functional Requirements

ID	Requirement	Priority
FR-9.2	Show current verification st	High
FR-9.3	Document submission flow	High
FR-9.4	Admin approval workflow	High

4. API / Data Requirements

Item	Status	Details
Document Upload	Missing	UI only
BVN Verification	Missing Backend	No integration
Liveness Check	Missing	Not implemented
Admin Approval	Mocked	Status updates only

PRD — Profile & Security

1. Overview

User profile management, security settings (password, PIN, 2FA), bank account management, and app preferences.

2. User Stories

- US-10.1: Edit Profile
- US-10.2: Change Password
- US-10.3: Update Transaction PIN
- US-10.4: Set Up Two-Factor Authentication
- US-10.5: Change App Theme

3. Functional Requirements

ID	Requirement	Priority
FR-10.2	Change password with validat	High
FR-10.3	Update transaction PIN (3-st	High
FR-10.4	2FA setup with QR code	Medium
FR-10.5	Theme switching (light/dark/	High
FR-10.6	Bank account management	Medium

4. API / Data Requirements

Item	Status	Details
Password Change	Mocked	No backend validation
PIN Update	Mocked	Hardcoded 1234
2FA Setup	Mocked	Static QR code
Theme	Implemented	Working with localStorage
Bank Management	Mocked	localStorage

PRD — Notifications

1. Overview

In-app notification system for users and admin push notification composer.

2. User Stories

- US-11.1: View Notifications

3. Functional Requirements

ID	Requirement	Priority
FR-11.2	Admin compose and send notif	High
FR-11.3	Notification history (admin)	Medium

4. API / Data Requirements

Item	Status	Details
Push Delivery	Missing Backend	No delivery system
Admin Composer	Mocked	Records to localStorage
Read/Unread Tracking	Missing	Not implemented

PRD — Admin Dashboard

1. Overview

Platform management interface for monitoring users, transactions, KYC, referrals, gift card orders, and system settings.

2. Goals & Objectives

- Full visibility into platform operations
- Efficient user and transaction management
- Configurable system parameters

3. User Personas

Persona	Description
Support Agent	(Intended) Limited access fo

4. User Stories

- US-12.1: View Admin Overview
- US-12.2: Manage Customers
- US-12.3: Monitor Transactions
- US-12.4: Approve/Reject Gift Card Orders
- US-12.5: Manage Referrals
- US-12.6: Process KYC Approvals
- US-12.7: Send Push Notifications
- US-12.8: Configure System Settings

5. Functional Requirements

ID	Requirement	Priority
FR-12.2	Customer list with filtering	High
FR-12.3	Customer detail view	High
FR-12.4	Transaction monitoring with	High
FR-12.5	Gift card order approval/rej	High
FR-12.6	Referral withdrawal processi	High
FR-12.7	KYC approval queue	High

ID	Requirement	Priority
FR-12.8	Push notification composer	Medium
FR-12.9	System settings (fees, limit	High

6. Non-Functional Requirements

- Dashboard loads within 2 seconds
- Charts render smoothly with 30+ data points
- Actions (approve/reject) are atomic and auditable (intended)

7. Edge Cases

Scenario	Expected Behavior
Concurrent admin actions	Prevent double-processing
Invalid configuration values	Show validation errors

8. API / Data Requirements

Item	Status	Details
Customer Data	Mocked	10 mock customers
Transaction Data	Mocked	10 mock orders
Revenue Chart	Mocked	30 days of mock data
Gift Card Orders	Mocked	localStorage store
KYC Queue	Mocked	Static list
Settings Storage	Mocked	localStorage
Admin Auth	Missing	No guards, anyone can access
Audit Trail	Missing	No action logging
Role Management	Missing	Single hardcoded admin

9. UI/UX Notes

- Sidebar navigation with sections
- Stats cards with sparkline charts
- Recharts area chart for revenue
- Expandable customer cards
- Tab-based filtering throughout

10. Metrics

Metric	Target
Data accuracy	100%
Dashboard load time	< 2s

INTENDED vs CURRENT (Admin)

Aspect	Intended	Current
Real-time Data	Live from database	Static mock data
Action Execution	Real effects on users/data	Mock status changes
Audit Trail	Full action logging	None
Multi-role Support	Super Admin, Support, Analys	Single hardcoded role
Fraud Detection	Automated flagging	None
Export/Reports	CSV/PDF generation	None

Celler — System Architecture Overview

1. Project Classification

Type: Frontend Prototype (Single Page Application)
Intended: Full-stack Crypto Exchange Platform
Current Reality: 100% client-side, zero backend

2. Current Architecture



Technology Stack

Layer	Technology
Framework	React 18.3.1
Language	TypeScript 5.8.3
Styling	Tailwind CSS 3.4.17
UI Components	shadcn/ui (Radix UI primitiv
Routing	React Router 6.30.1
State	React Context + localStorage
Forms	React Hook Form + Zod
Charts	Recharts 2.15.4
Icons	Lucide React 0.462.0
Testing	Vitest 3.2.4 + Testing Libra
Package Manager	Bun (bun.lock) / npm

3. Intended Production Architecture





4. Route Architecture

Public Routes (No Auth)

Route	Component	Auth Required
/welcome	Welcome	No
/signin	SignIn	No
/signup	SignUp	No

Authenticated User Routes

Route	Component	Auth Required
/swap	Swap	Yes
/history	History	Yes
/profile	Profile	Yes
/coin/:coinId	CoinDetail	Yes
/notifications	Notifications	Yes
/withdraw/:coinId	Withdraw	Yes
/receive/:coinId	Receive	Yes
/manage-assets	ManageAssets	Yes
/transaction/:txId	TransactionDetail	Yes
/receipt/:txId	Receipt	Yes
/referral	Referral	Yes
/bank-details	BankDetails	Yes
/kyc	KYC	Yes
/update-password	UpdatePassword	Yes
/update-pin	UpdatePin	Yes
/two-factor	TwoFactor	Yes
/appearance	Appearance	Yes
/support	Support	Yes
/edit-profile	EditProfile	Yes
/giftcards	Giftcards	Yes
/giftcard/sell/:brandId	GiftcardSell	Yes
/giftcard-orders	GiftcardOrders	Yes

Admin Routes

Route	Component	Auth Required
/admin/customers	AdminCustomers	No (should be Yes)
/admin/customers/:customerI	AdminCustomerDetail	No (should be Yes)
/admin/orders	AdminTransactions	No (should be Yes)
/admin/transactions	AdminTransactions	No (should be Yes)
/admin/giftcard-orders	AdminGiftcardOrders	No (should be Yes)
/admin/referrals	AdminReferrals	No (should be Yes)
/admin/kyc	AdminKYC	No (should be Yes)
/admin/notifications	AdminPushNotifications	No (should be Yes)
/admin/settings	AdminSettings	No (should be Yes)

404

Route	Component
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5. Data Flow

Current (Mock)

User Action → Component → localStorage Store → State Update → Re-render

Intended (Production)

User Action → Component → TanStack Query mutation → API → Service → Database

↓

User Action ← Component ← TanStack Query cache ← API ← Service

6. State Management Architecture

Store	Location	Data	Persistence
ThemeContext	`src/contexts/ThemeContext.t	Theme mode	localStorage (cex_theme)
Crypto Store	src/lib/crypto.ts	Wallet, transactions, coins,	localStorage (cex_wallet,
Referral Store	src/lib/referral.ts	Referrals, withdrawals, conf	localStorage (`cex_referrals
Giftcard Store	src/data/giftcardData.ts	Orders	localStorage (`cex_giftcard_

Store	Location	Data	Persistence
Fee Store	src/data/feeConfig.ts	Fee percentages	localStorage ('cex_fee_conf')
KYC Config	src/data/kycConfig.ts	Tier limits	localStorage ('cex_kyc_conf')

7. Third-Party Dependencies

Active Dependencies

Package	Purpose
lucide-react	Icon library
@radix-ui/*	Headless UI primitives
react-hook-form + zod	Form validation
recharts	Charts (admin dashboard)
sonner	Toast notifications
input-otp	OTP input
class-variance-authority	Component variants
clsx + tailwind-merge	CSS class merging

Dead/Unused Dependencies

Package	Status	Notes
next-themes	Unused	Custom ThemeContext used ins
embla-carousel-react	Unused	shadcn/ui carousel not activ
cmdk	Unused	Command palette not implemen
react-day-picker	Unused	Calendar not used in any pag
vaul	Unused	Drawer not used
date-fns	Minimal	Imported but rarely used

8. Security Architecture (Current vs Intended)

Aspect	Current	Intended
Authorization	None	Role-based access control
Password Storage	Plaintext in localStorage	bcrypt/argon2 hash
Session Management	Boolean flag	JWT with expiry
PIN Storage	Hardcoded 1234	Hashed, user-set
API Security	N/A (no API)	HTTPS, rate limiting, CORS
Data Encryption	None	TLS in transit, encryption a
Admin Access	No guards	Separate admin auth + RBAC

Celler — User Stories

EPIC 1: Authentication & Onboarding

US-1.1 — View Splash Screen

As a first-time visitor,
I want to see a branded splash screen when I open the app,
so that I know I'm using the Celler platform.

Acceptance Criteria:

Given I open the Celler app
When the splash screen loads
Then I see the Celler logo with gradient branding
And I am automatically redirected after 2.5 seconds

US-1.2 — View Welcome Carousel

As a new visitor,
I want to see an onboarding carousel explaining the app's value,
so that I understand what Celler offers before signing up.

Acceptance Criteria:

Given I have passed the splash screen
When I land on the welcome page
Then I see a 3-slide carousel with value propositions
And I can navigate between slides
And I see "Login" and "Sign Up" CTA buttons

US-1.3 — Sign In to Existing Account

As a registered user,
I want to sign in with my email and password,
so that I can access my wallet and perform transactions.

Acceptance Criteria:

Given I am on the sign-in page
When I enter my registered email and correct password
And I tap "Sign In"
Then I am redirected to the dashboard
And my session is persisted in localStorage

Given I enter an incorrect password
When I tap "Sign In"
Then I see an error message indicating invalid credentials

US-1.4 — Create New Account

As a new user,
I want to create an account through a multi-step process,
so that I can start using the platform securely.

Acceptance Criteria:

Given I am on the sign-up page
When I enter my email address
And I proceed to the OTP step
And I enter any valid 6-digit code
Then I proceed to the password creation step

Given I am on the password creation step
When I enter a password with 8+ characters, uppercase, and special character
And I confirm the password matches
Then my account is created
And I see a success confirmation
And I am redirected to sign-in or dashboard

US-1.5 — Stay Logged In Across Sessions

As a returning user,
I want to remain logged in when I reopen the app,
so that I don't have to sign in every time.

Acceptance Criteria:

Given I have previously logged in
When I close and reopen the app
Then I am automatically redirected to the dashboard
And I do not see the sign-in screen

US-1.6 — Log Out

As a logged-in user,
I want to log out of my account,
so that my account is secure when I'm done using the app.

Acceptance Criteria:

Given I am logged in and on the profile page
When I tap "Log Out"
Then my session is cleared from localStorage
And I am redirected to the sign-in page

EPIC 2: Wallet & Dashboard

US-2.1 — View Portfolio Balance

As a logged-in user,
I want to see my total portfolio balance in USD and NGN,
so that I know the total value of my assets at a glance.
Acceptance Criteria:

Given I am on the dashboard
When the page loads
Then I see my total balance displayed in USD
And I see the equivalent NGN value
And the balance reflects all coins in my wallet

US-2.2 — View Asset List

As a user,
I want to see a list of my cryptocurrency holdings,
so that I can monitor each asset's balance and value.
Acceptance Criteria:

Given I am on the dashboard
When I scroll to the asset list
Then I see each coin with its icon, name, balance, and USD value
And I see the 24h price change percentage
And coins are sorted by default visibility settings

US-2.3 — Access Quick Actions

As a user,
I want quick action buttons for common operations,
so that I can navigate to key features without searching.
Acceptance Criteria:

Given I am on the dashboard
When I view the quick actions section
Then I see buttons for Buy, Sell, Send, Receive, Swap, and More
And tapping each button navigates to the corresponding feature

US-2.4 — Manage Visible Assets

As a user,
I want to toggle which coins appear on my dashboard,
so that I can customize my view to show only relevant assets.
Acceptance Criteria:

Given I am on the dashboard
When I tap "Manage Assets"
Then I see a list of all supported coins with toggle switches
And toggling a coin on/off updates my dashboard view
And my preferences are saved to localStorage

US-2.5 — View Coin Details

As a user,
I want to tap on a coin to see its detailed information,
so that I can perform specific actions for that asset.

Acceptance Criteria:

Given I am on the dashboard
When I tap on a specific coin
Then I am taken to the coin detail page
And I see the coin's balance, USD value, and quick action buttons
And I see the transaction history for that specific coin

EPIC 3: Crypto Swap

US-3.1 — Swap Cryptocurrency

As a user,
I want to exchange one cryptocurrency for another,
so that I can rebalance my portfolio or acquire a different asset.

Acceptance Criteria:

Given I am on the swap page
When I select a "from" coin and enter an amount
Then I see the calculated "to" amount based on current rates
And I see the applicable fee and slippage tolerance
And I can review the details before confirming
And upon confirmation, my wallet balances are updated
And I see a success animation with transaction timeline

US-3.2 — View Swap Rate and Fees

As a user,
I want to see the exchange rate and fees before confirming a swap,
so that I can make an informed decision.

Acceptance Criteria:

Given I have selected coins and entered an amount
When I view the swap summary
Then I see the exchange rate (toCoin/fromCoin)
And I see the fee percentage and amount
And I see the slippage tolerance

US-3.3 — Swap Coin Selection

As a user,
I want to browse and select coins from a list for swapping,
so that I can choose any supported pair.

Acceptance Criteria:

Given I am on the swap page

When I tap the coin selector
 Then I see a searchable list of all supported coins
 And I can select any coin as the "from" or "to" asset
 And I cannot select the same coin for both sides

EPIC 4: Withdrawal

US-4.1 — Withdraw Cryptocurrency

As a user,
 I want to send cryptocurrency to an external wallet address,
 so that I can transfer my assets off the platform.

Acceptance Criteria:

Given I am on the withdrawal page for a crypto coin
 When I enter a withdrawal amount
 And I enter a valid recipient address
 And I review the summary including fees
 And I authenticate with FaceID or PIN
 Then the withdrawal is processed
 And my wallet balance is reduced
 And I see a success confirmation
 And the transaction appears in my history

US-4.2 — Withdraw NGN to Bank

As a user,
 I want to withdraw Nigerian Naira to my bank account,
 so that I can access my fiat funds.

Acceptance Criteria:

Given I am on the NGN withdrawal page
 When I enter a withdrawal amount
 And I select a linked bank account
 And I review the summary including the flat 100 NGN fee
 And I authenticate with FaceID or PIN
 Then the withdrawal is processed
 And my NGN balance is reduced
 And I see a success confirmation

US-4.3 — Add Bank Account

As a user,
 I want to add a bank account for NGN withdrawals,
 so that I have a destination for my fiat withdrawals.

Acceptance Criteria:

Given I am on the bank details page
 When I enter bank name, account number, and account name
 And I save the account

Then the bank account is added to my list
And I can set it as default

US-4.4 — Authenticate Withdrawal

As a user,
I want to authenticate before completing a withdrawal,
so that unauthorized withdrawals are prevented.

Acceptance Criteria:

Given I am on the withdrawal review step
When I choose FaceID authentication
Then I see a simulated FaceID scan for 2 seconds
And upon completion, the withdrawal proceeds

Given I choose PIN authentication
When I enter the correct 4-digit PIN
Then the withdrawal proceeds
And an incorrect PIN shows an error

EPIC 5: Receive & Deposit

US-5.1 — View Crypto Deposit Address

As a user,
I want to see my deposit address and QR code for a cryptocurrency,
so that I can receive funds from external wallets.

Acceptance Criteria:

Given I am on the receive page for a crypto coin
When the page loads
Then I see a QR code representing my deposit address
And I see the address as text
And I can copy the address to clipboard

US-5.2 — View NGN Deposit Details

As a user,
I want to see bank details for depositing NGN,
so that I can fund my account via bank transfer.

Acceptance Criteria:

Given I am on the NGN receive page
When the page loads
Then I see the deposit bank name, account number, and account name
And I can copy the account number to clipboard

EPIC 6: Transaction History

US-6.1 — View Transaction History

As a user,

I want to see a list of all my transactions,
so that I can track my activity on the platform.

Acceptance Criteria:

Given I am on the history page
When the page loads
Then I see a chronological list of transactions
And each transaction shows type, coin, amount, date, and status
And transactions are grouped or sortable by date

US-6.2 — Filter Transactions

As a user,

I want to filter transactions by type and status,
so that I can find specific transactions quickly.

Acceptance Criteria:

Given I am on the history page
When I select a transaction type filter
Then only transactions of that type are shown
When I select a status filter
Then only transactions with that status are shown

US-6.3 — Search Transactions

As a user,

I want to search my transactions,
so that I can find a specific transaction by keyword.

Acceptance Criteria:

Given I am on the history page
When I enter a search term
Then transactions matching the term are filtered and displayed

US-6.4 — View Transaction Details

As a user,

I want to tap on a transaction to see full details,
so that I can review all information about that transaction.

Acceptance Criteria:

Given I am on the history page
When I tap a transaction
Then I see a detail page with type, amount, fees, date, status, and addresses

And I see a calculation breakdown for swaps

US-6.5 — View and Share Receipt

As a user,

I want to generate a receipt for a transaction,
so that I can share or print proof of the transaction.

Acceptance Criteria:

Given I am viewing a transaction detail
When I tap "View Receipt"
Then I see a formatted receipt with zigzag card design
And the receipt includes all transaction details
And I can share or print the receipt

EPIC 7: Gift Card Trading

US-7.1 — Browse Gift Card Brands

As a user,

I want to browse available gift card brands,
so that I can find the brand I want to sell.

Acceptance Criteria:

Given I am on the gift cards page
When the page loads
Then I see a grid of gift card brands organized by category
And each brand shows its logo and name

US-7.2 — Sell a Gift Card

As a user,

I want to sell my gift card for NGN,
so that I can convert unused gift cards to cash.

Acceptance Criteria:

Given I am on the gift card sell page for a brand
When I enter the gift card amount
And I select the denomination tier
Then I see the calculated NGN payout based on the rate
And I can upload a card image
And I can review the order details
And upon submission, the order is created with "pending" status
And I see a success confirmation

US-7.3 — View Gift Card Orders

As a user,
I want to see my gift card order history,
so that I can track the status of my sales.
Acceptance Criteria:

Given I am on the gift card orders page
When the page loads
Then I see a list of my gift card orders
And each order shows brand, amount, NGN payout, and status
And I can see pending, approved, and rejected orders

EPIC 8: Referral Program

US-8.1 — View Referral Code and Link

As a user,
I want to see my unique referral code and shareable link,
so that I can invite others to join Celler.
Acceptance Criteria:

Given I am on the referral page
When the page loads
Then I see my unique referral code
And I see a shareable referral link
And I can copy both to clipboard

US-8.2 — Track Referral Earnings

As a user,
I want to see my referral earnings and statistics,
so that I know how much I've earned and can earn.
Acceptance Criteria:

Given I am on the referral page
When I view my earnings section
Then I see total earned, total withdrawn, and available balance
And I see my referral list with their status (signed up, verified, traded)

US-8.3 — Withdraw Referral Earnings

As a user,
I want to request a withdrawal of my referral earnings,
so that I can access my earned rewards.
Acceptance Criteria:

Given I have available referral earnings
When I request a withdrawal
Then the request is submitted with "pending" status
And I cannot exceed the weekly withdrawal limit of 10,000 NGN

And the withdrawal requires admin approval

EPIC 9: KYC Verification

US-9.1 — View KYC Tiers and Limits

As a user,
I want to see the KYC tiers and their associated limits,
so that I understand what verification level I need.
Acceptance Criteria:

Given I am on the KYC page
When the page loads
Then I see all 3 KYC tiers with their requirements and limits
And I see my current verification status

US-9.2 — Complete KYC Verification

As a user,
I want to submit KYC documents for verification,
so that I can increase my transaction limits.
Acceptance Criteria:

Given I am on the KYC page
When I select a tier to verify
And I submit the required documents
Then my verification is submitted for review
And I see the expected approval type (auto or manual)

EPIC 10: Profile & Security

US-10.1 — Edit Profile

As a user,
I want to update my profile information,
so that my account details are current.
Acceptance Criteria:

Given I am on the edit profile page
When I update my full name, username, or phone number
And I save the changes
Then my profile is updated
And the changes are reflected across the app

US-10.2 — Change Password

As a user,
I want to change my account password,
so that I can maintain account security.

Acceptance Criteria:

Given I am on the update password page
When I enter my current password and a new valid password
And I confirm the new password
Then my password is updated

US-10.3 — Update Transaction PIN

As a user,
I want to change my transaction PIN,
so that I can keep my transactions secure.

Acceptance Criteria:

Given I am on the update PIN page
When I enter my current PIN, a new PIN, and confirm the new PIN
And the new PINs match
Then my transaction PIN is updated

US-10.4 — Set Up Two-Factor Authentication

As a user,
I want to enable 2FA on my account,
so that I add an extra layer of security.

Acceptance Criteria:

Given I am on the 2FA setup page
When I view the QR code and secret key
And I enter the verification code from my authenticator app
Then 2FA is enabled on my account

US-10.5 — Change App Theme

As a user,
I want to switch between light, dark, and system themes,
so that the app matches my visual preference.

Acceptance Criteria:

Given I am on the appearance page
When I select a theme option
Then the app theme changes immediately
And my preference is saved to localStorage

EPIC 11: Notifications

US-11.1 — View Notifications

As a user,
I want to see my in-app notifications,
so that I stay informed about platform updates and my account.
Acceptance Criteria:

Given I am on the notifications page
When the page loads
Then I see a list of notifications with title, description, and timestamp
And notifications include types like rate updates, new features, and price alerts

EPIC 12: Admin Dashboard

US-12.1 — View Admin Overview

As an admin,
I want to see a dashboard with key platform metrics,
so that I can monitor the health of the platform.
Acceptance Criteria:

Given I am on the admin dashboard
When the page loads
Then I see total registered users, verified users, active users
And I see total transactions and volume
And I see a revenue chart over time
And I see recent orders

US-12.2 — Manage Customers

As an admin,
I want to view and manage platform customers,
so that I can monitor user activity and status.
Acceptance Criteria:

Given I am on the customers page
When I view the customer list
Then I see customers with their KYC level, status, and trade counts
And I can filter by tabs (all, verified, unverified)
And I can tap a customer to see their detailed profile

US-12.3 — Monitor Transactions

As an admin,
I want to view all platform transactions,
so that I can monitor trading activity.

Acceptance Criteria:

Given I am on the orders/transactions page
When I view the transaction list
Then I see all orders with type, amount, user, and status
And I can filter by transaction type tabs

US-12.4 — Approve/Reject Gift Card Orders

As an admin,
I want to review and process gift card orders,
so that I can approve legitimate sales and reject fraudulent ones.

Acceptance Criteria:

Given I am on the gift card orders page
When I view pending orders
Then I see order details including brand, amount, and card image
And I can approve or reject each order
And approving credits the NGN payout to the user's wallet

US-12.5 — Manage Referrals

As an admin,
I want to review and process referral withdrawal requests,
so that I can control reward payouts.

Acceptance Criteria:

Given I am on the admin referrals page
When I view pending withdrawal requests
Then I can approve or reject each request
And I can view all referrals and their status
And I can update referral configuration (reward amount, weekly limit)

US-12.6 — Process KYC Approvals

As an admin,
I want to review and approve KYC submissions,
so that I can verify user identities for higher tiers.

Acceptance Criteria:

Given I am on the admin KYC page
When I view pending approvals
Then I see the list of users awaiting verification
And I can approve or reject each submission
And I see a verification log of past actions

US-12.7 — Send Push Notifications

As an admin,

I want to compose and send push notifications to users,
so that I can communicate important updates.

Acceptance Criteria:

Given I am on the push notifications page
When I compose a notification with title, message, and audience
And I send it
Then the notification is recorded in the history
And it is marked with the sent date and recipient count

US-12.8 — Configure System Settings

As an admin,

I want to adjust platform-wide settings,
so that I can control fees, limits, and referral parameters.

Acceptance Criteria:

Given I am on the admin settings page
When I view the settings tabs
Then I can adjust trade fees, withdrawal fees, swap fees, and NGN rate markup
And I can update KYC tier limits
And I can update referral reward amount and weekly withdrawal limit
And changes are saved and reflected in the user app

Story Summary

Epic	Story Count	Stories
Wallet & Dashboard	5	US-2.1 to US-2.5
Crypto Swap	3	US-3.1 to US-3.3
Withdrawal	4	US-4.1 to US-4.4
Receive & Deposit	2	US-5.1 to US-5.2
Transaction History	5	US-6.1 to US-6.5
Gift Card Trading	3	US-7.1 to US-7.3
Referral Program	3	US-8.1 to US-8.3
KYC Verification	2	US-9.1 to US-9.2
Profile & Security	5	US-10.1 to US-10.5
Notifications	1	US-11.1
Admin Dashboard	8	US-12.1 to US-12.8
Total	47	

Celler — QA & QC Test Cases

Date: April 3, 2026

Version: 0.1.0 (Prototype)

Test Environment: Browser (Chrome, Safari, Firefox)

Note: All tests are against the frontend prototype with mocked data. Backend-dependent tests are marked as Pending.

1. Authentication Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
AUTH-02	First-time auto-logout	High	No localStorage data	1. Clear localStorage	Auto-creates mock user	Pending	
AUTH-03	Sign in with valid credentials	High	Mock user exists	1. Navigate to /signin	Redirects to /dashboard	Pending	
AUTH-04	Sign in with wrong credentials	High	Mock user exists	1. Navigate to /signin	Shows error message	Pending	
AUTH-05	Sign in with empty fields	Medium	On /signin	1. Leave fields empty	Shows validation errors	Pending	
AUTH-06	Sign up — email step	High	On /signup	1. Enter valid email	Proceeds to OTP step	Pending	
AUTH-07	Sign up — OTP step	High	On OTP step	1. Enter any 6-digit code	Proceeds to password step	Pending	
AUTH-08	Sign up — OTP verification	Medium	On OTP step	1. Enter 5-digit code	Continue button disabled	Pending	
AUTH-09	Sign up — weak password	High	On password step	1. Enter "password123"	Shows password requirements	Pending	
AUTH-10	Sign up — password strength	High	On password step	1. Enter valid password	Shows "Passwords match"	Pending	
AUTH-11	Sign up — valid flow	High	On /signup	1. Complete all 4 steps	Account created, success message	Pending	
AUTH-12	Session persistence	High	Logged in	1. Log in 2. Close browser	Automatically logged in on return	Pending	
AUTH-13	Logout	High	Logged in	1. Go to Profile 2. Click Logout	Session cleared, redirected to /signin	Pending	
AUTH-14	Auth guard redirection	High	Not logged in	1. Clear localStorage	Redirects to /signin	Pending	

2. Wallet & Dashboard Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
WAL-02	Asset list displays correctly	High	Logged in	1. Navigate to /dashboard	All coins with balance	Pending	
WAL-03	Quick action buttons	High	On dashboard	1. Tap each quick action	Each navigates to correct page	Pending	
WAL-04	Quick action — Buy	High	On dashboard	1. Tap "Buy" button	Should navigate to Buy page	Expected Fail	
WAL-05	Quick action — Sell	High	On dashboard	1. Tap "Sell" button	Should navigate to Sell page	Expected Fail	
WAL-06	Manage assets — Toggle	Medium	On dashboard	1. Tap "Manage Assets"	Toggled coin no longer visible	Pending	
WAL-07	Manage assets — Persistence	High	On dashboard	1. Toggle coins 2. Refresh page	Preferences persist	Pending	
WAL-08	Coin detail navigation	High	On dashboard	1. Tap on BTC coin	Navigates to /coin/BTC	Pending	
WAL-09	Coin detail — buy/sell price	Medium	On /coin/BTC	1. Tap Buy or Sell button	Currently 404	Expected Fail	
WAL-10	Promo banner carousel	Low	On dashboard	1. View promo banners	3 banners rotate on click	Pending	
WAL-11	Mobile bottom navigation	High	Mobile viewport	1. Resize to < 768px	Bottom tab navigation visible	Pending	
WAL-12	Desktop sidebar navigation	High	Desktop viewport	1. Resize to > 768px	Left sidebar visible	Pending	

3. Swap Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
SWAP-02	Coin selection	High	On /swap	1. Tap from coin se	ETH shown as fron	Pending	
SWAP-03	Rate calculation	High	On /swap	1. Select BTC→ETH	Rate of BTC→ETH am	Pending	
SWAP-04	Fee display	Medium	On /swap	1. Enter amount	Fee amount and pe	Pending	
SWAP-05	Insufficient balance	High	On /swap	1. Enter amount ex	Error shown, confir	Disabl	
SWAP-06	Same coin selectio	Medium	On /swap	1. Select BTC for b	Prevented or show	Pending	
SWAP-07	Swap confirmation	High	On /swap	1. Enter valid swap	3-sec animation, R	Pending	
SWAP-08	Wallet update after	High	After swap	1. Check dashboar	From coin decreas	Pending	
SWAP-09	Transaction record	High	After swap	1. Check history	New swap transaction	Appears	
SWAP-10	Zero amount	Low	On /swap	1. Enter 0 amount	Confirm disabled	Pending	

4. Withdrawal Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
WDR-02	Amount entry with	High	On withdrawal	1. Enter amount us	Amount displayed	Pending	
WDR-03	Crypto address input	High	On withdrawal	1. Enter crypto add	Address accepted,	Pending	
WDR-04	Bank selection for	High	On /withdraw/NGN	1. Select linked bar	Bank details show	Pending	
WDR-05	Fee display	Medium	On withdrawal	1. Enter amount	Fee shown (NGN: 100	Pending	
WDR-06	Review summary	High	On withdrawal	1. Complete amou	Full summary with	Pending	
WDR-07	FaceID authentication	Medium	On auth step	1. Select FaceID	2-second scan anim	Pending	
WDR-08	PIN authentication	High	On auth step	1. Select PIN 2	Withdrawal procee	Pending	
WDR-09	PIN authentication	High	On auth step	1. Select PIN 2	Error shown, retry a	Pending	
WDR-10	Insufficient balance	High	On withdrawal	1. Enter amount >	Error shown, cannc	Pending	
WDR-11	Success animation	Medium	After auth	1. Complete withdra	Animated timeline w	Pending	
WDR-12	Wallet balance upd	High	After withdrawal	1. Check dashboar	Balance reduced b	Pending	

5. Receive & Deposit Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
RCV-02	Copy address	Medium	On /receive/BTC	1. Tap copy button	Address copied to	Pending	
RCV-03	NGN receive page	High	Logged in	1. Navigate to /rece	Bank details display	Pending	
RCV-04	Copy account num	Medium	On /receive/NGN	1. Tap copy on acc	Number copied to	Pending	
RCV-05	Network selection	Low	On /receive/ETH	1. View network opt	Network info display	Pending	

6. Transaction History Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
TXN-02	Filter by type	High	On /history	1. Select "swap" filter	Only swap transactions	Pending	
TXN-03	Filter by status	Medium	On /history	1. Select "completed" filter	Only completed transactions	Pending	
TXN-04	Search transaction	Medium	On /history	1. Enter search term	Matching transactions	Pending	
TXN-05	Transaction detail	High	On /history	1. Tap a transaction	Detail page with full info	Pending	
TXN-06	Receipt view	Medium	On transaction detail	1. Tap "View Receipt"	Receipt with zigzag	Pending	
TXN-07	Empty state	Low	No transactions	1. Clear all transactions	Empty state with CTA	Pending	

7. Gift Card Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
GC-02	Select brand	High	On /giftcards	1. Tap Amazon brand	Navigates to sell page	Pending	
GC-03	Amount and tier selection	High	On sell page	1. Enter amount 2. Select tier	NGN payout calculation	Pending	
GC-04	Payout calculation	High	On sell page	1. Enter \$100 Amount	$NGN = 100 \times 0.70$	Pending	
GC-05	Order submission	High	On sell page	1. Complete all steps	Order created with "Pending" status	Pending	
GC-06	Order history	High	After order	1. Navigate to /giftcard-history	Order appears in list	Pending	
GC-07	Admin approval flow	High	Admin access	1. Go to /admin/giftcard-approval	Order status changes to "Approved"	Pending	
GC-08	NGN payout on approval	High	After admin approval	1. Check user NGN balance	NGN increased by payout	Pending	

8. Referral Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
REF-02	Copy referral code	Medium	On /referral	1. Tap copy on code	Code copied to clipboard	Pending	
REF-03	Copy referral link	Medium	On /referral	1. Tap copy on link	Link copied to clipboard	Pending	
REF-04	Withdrawal request	High	On /referral	1. Request withdrawal	Request submitted	Pending	
REF-05	Weekly limit enforcement	High	On /referral	1. Request withdrawal 2. Exceed limit	Request shown, request blocked	Pending	

9. KYC Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
KYC-02	Tier limits display	Medium	On /kyc	1. View each tier	Correct limits shown	Pending	
KYC-03	KYC submission	Medium	On /kyc	1. Select tier 2. Submit	Submission recorded	Pending	

10. Profile & Security Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
PROF-02	Edit profile	Medium	On /edit-profile	1. Update name/us	Changes saved an	Pending	
PROF-03	Update password	Medium	On /update-passwor	1. Enter current + n	Password updated	Pending	
PROF-04	Update PIN	High	On /update-pin	1. Enter current PIN	PIN updated	Pending	
PROF-05	Update PIN — mis	Medium	On /update-pin	1. Enter different co	Error Shown	Pending	
PROF-06	2FA setup	Medium	On /two-factor	1. View QR code a	QR code and secre	Pending	
PROF-07	Theme switching	High	On /appearance	1. Select dark/light	Theme changes im	Pending	
PROF-08	Theme persistence	Medium	After theme change	1. Change theme<	Theme persists	Pending	
PROF-09	Bank account mana	Medium	On /bank-details	1. Add new bank ac	Account added to l	Pending	

11. Admin Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
ADM-02	Customer list	High	On /admin	1. Navigate to /adn	Customer list with t	Pending	
ADM-03	Customer detail	High	On customers	1. Tap a customer	Detail page with pro	Pending	
ADM-04	Transaction monit	High	On /admin	1. Navigate to /adn	All orders with type	Pending	
ADM-05	Gift card order app	High	On /admin/giftcard-	1. View pending ord	Status changes, N	Pending	
ADM-06	Referral managem	Medium	On /admin/referrals	1. View withdrawal	Status changes to	Pending	
ADM-07	KYC approvals	High	On /admin/kyc	1. View pending KY	KYC status updated	Pending	
ADM-08	Push notification c	Medium	On /admin/notificat	1. Compose notific	Recorded in history	Pending	
ADM-09	System settings —	High	On /admin/settings	1. Update fee perce	Changes saved to	Pending	
ADM-10	System settings —	Medium	On /admin/settings	1. Update KYC tier	Changes reflected	Pending	
ADM-11	System settings —	Medium	On /admin/settings	1. Update reward a	Changes reflected	Pending	
ADM-12	Admin access with	Critical	Not logged in	1. Navigate to /adm	Currently accessibl	Expected Pass (as bug)	

12. Notifications Module

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
NOT-02	Notification conten	Low	On /notifications	1. View each notific	Title, description, ti	Pending	

13. Edge Cases & Error Handling

TC-ID	Title	Priority	Preconditions	Test Steps	Expected Result	Actual	Status
EDGE-02	External image fail	Medium	Any page	1. Block cryptologo	Fallback placeholder	Pending	
EDGE-03	localStorage cleared	High	Logged in	1. Clear localStorage	Reset as new user	Pending	
EDGE-04	Rapid navigation	Low	Logged in	1. Rapidly click nav	No crashes or race	Pending	
EDGE-05	Mobile viewport	High	Any page	1. Resize to 375px	Responsive layout,	Pending	
EDGE-06	Desktop viewport	High	Any page	1. Resize to 1440p	Sidebar layout, no	Pending	

14. Smoke Test Suite

Run these tests on every build to verify core functionality:

#	Test	Expected
2	Sign in with demo@celler.app	Dashboard loads
3	Dashboard shows balance	Balance > 0
4	Navigate to /swap	Swap page loads
5	Navigate to /history	Transaction list loads
6	Navigate to /profile	Profile page loads
7	Navigate to /giftcards	Gift card grid loads
8	Navigate to /referral	Referral page loads
9	Navigate to /kyc	KYC page loads
10	Navigate to /admin	Admin dashboard loads
11	Logout	Redirects to /signin
12	404 page	/nonexistent shows 404

15. Regression Test Scenarios

#	Scenario	Priority	Description
REG-02	Balance accuracy	High	Verify all wallet operations
REG-03	Theme persistence	Medium	Verify theme persists across
REG-04	Session persistence	High	Verify login persists across
REG-05	Gift card rate calculation	High	Verify payout calculation is
REG-06	Referral weekly limit	High	Verify weekly limit enforcem
REG-07	Admin settings propagation	Medium	Verify admin fee changes ref

Celler — Gap Analysis

Date: April 3, 2026

Version: 0.1.0 (Prototype)

1. Missing Features

ID	Feature	Severity	Description	Impact
G-2	Real Authentication	Critical	No real auth server, JWT, or	No security, no user isolation
G-3	Admin Authentication	Critical	Admin routes have no guards	Complete security breach
G-4	Buy Flow	High	Routes /buy and /sell no	Broken user journey
G-5	Sell Flow	High	Same as G-4	Broken user journey
G-6	Real Price Feeds	High	All crypto prices are hardcoded	No real market data
G-7	Blockchain Integration	High	No wallet addresses, no real	Core feature non-functional
G-8	Payment Gateway	High	No NGN deposit/withdrawal pr	Fiat operations non-function
G-9	Email/SMS Service	High	No real OTP delivery, no ema	Auth and notifications non-f
G-10	Document Upload	High	KYC document upload is UI on	KYC non-functional
G-11	Image Upload	High	Gift card image upload is UI	Gift card selling non-functi
G-12	Forgot Password	Medium	Link exists but page not imp	Users cannot recover account
G-13	Address Validation	Medium	No crypto address format che	Invalid addresses accepted
G-14	Withdrawal Limits	Medium	KYC tier limits not enforced	Users can exceed limits
G-15	Pagination	Medium	Transaction history has no p	Performance issues at scale
G-16	Export/Reports	Medium	No CSV/PDF export for transa	Limited auditability
G-17	Audit Trail	Medium	No admin action logging	No accountability
G-18	Multi-role Admin	Medium	Single hardcoded admin role	No separation of duties
G-19	Fraud Detection	Medium	No duplicate card detection,	Fraud risk
G-20	Portfolio Chart	Low	No historical balance visual	Reduced user experience
G-21	Bill Payments	Low	Banner shows "coming soon"	Missing feature
G-22	Live Chat Support	Low	Support page has static link	No real support channel
G-23	Push Notifications	Low	No real push notification de	Notifications are static
G-24	Read/Unread Tracking	Low	Notifications don't track re	Poor UX
G-25	Offline Support	Low	No caching or offline mode	App unusable without interne

2. Broken Flows

ID	Flow	Issue	Severity
B-2	Sell crypto from dashboard	Route /sell not defined →	High
B-3	Buy crypto from CoinDetail	Route /buy not defined → 4	High
B-4	Sell crypto from CoinDetail	Route /sell not defined →	High
B-5	Forgot password from sign-in	No page exists	Medium

ID	Flow	Issue	Severity
B-6	Index.tsx page	File exists but not imported	Low

3. Mocked/Placeholder Logic

ID	Component	What's Mocked	Real Implementation Ne
M-2	All coin prices	Hardcoded static values	Real-time price API (CoinGec
M-3	Swap execution	localStorage balance update	Real DEX/order book executio
M-4	Withdrawal processing	localStorage balance deducti	Blockchain transaction + pay
M-5	Receive addresses	Static hardcoded addresses	Real address generation per
M-6	Transaction history	8 pre-seeded mock transactio	Real transaction database
M-7	Gift card processing	localStorage order store	Real order processing + frau
M-8	Referral tracking	Mock data in localStorage	Real attribution system
M-9	KYC verification	UI only, no real verificatio	BVN check, ID verification,
M-10	Notifications	4 hardcoded notifications	Real notification delivery s
M-11	Admin data	All mock data in `adminMockD	Real database queries
M-12	FaceID auth	2-second timer	Real biometric API
M-13	2FA setup	Static QR code and secret	Real TOTP generation and ver
M-14	OTP verification	Any 6-digit code accepted	Real OTP generation and vali

4. Backend Gaps

Gap	Description	Priority
No Database	No PostgreSQL, MongoDB, or a	Critical
No Auth Service	No Supabase, Firebase, Auth0	Critical
No Blockchain Integration	No Web3.js, ethers.js, or no	Critical
No Payment Processor	No Paystack, Flutterwave, or	Critical
No File Storage	No S3, GCS, or file upload s	High
No Email Service	No SendGrid, Resend, or SMTP	High
No SMS Service	No Twilio, Termii, or SMS ga	High
No WebSocket	No real-time communication I	Medium
No Job Queue	No Redis, Bull, or backgroun	Medium
No Monitoring	No Sentry, Datadog, or error	Medium
No CDN	No CloudFront or image optim	Low

5. UX Inconsistencies

ID	Issue	Location	Description
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ID	Issue	Location	Description
UX-2	Title placeholder	index.html	Still says "Lovable App"
UX-3	OpenGraph tags	index.html	Generic placeholder content
UX-4	Dead dependency	package.json	next-themes installed but
UX-5	Unused QueryClient	main.tsx	TanStack Query set up but ne
UX-6	External image dependencies	CoinIcon.tsx, gift cards	Relies on cryptologos.cc and
UX-7	No loading states	Most pages	No skeleton screens for data
UX-8	No error boundaries	App.tsx	No React error boundary for

6. Classification Summary

Severity	Count	Examples
High	12	Buy/sell missing, no price f
Medium	8	No address validation, no pa
Low	8	No portfolio chart, no offli

Total Gaps Identified: 31

Celler — Technical Debt Analysis

Date: April 3, 2026

1. Structural Issues

ID	Issue	Location	Severity	Description	Recommended Fix
TD-2	No route guards for admin	App.tsx	Critical	Admin routes accessible to all	Add admin auth guard with router
TD-3	Monolithic store	lib/crypto.ts	High	Single file handles wallets, balances, etc.	Split into separate stores/modules
TD-4	No error boundaries	App.tsx	High	No React error boundary	Add ErrorBoundary wrapper
TD-5	Dead file	pages/Index.tsx	Low	Exists but not imported	Remove file

2. Hardcoded Values

ID	Value	Location	Severity	Description	Fix
TD-7	NGN_RATE 1410	lib/crypto.ts	High	Fixed exchange rate	Fetch from API or admin-config
TD-8	Coin prices	lib/crypto.ts	High	Static prices	Real-time price feed
TD-9	Swap fee 0.2%	pages/Swap.tsx	Medium	Differs from config (0.8%)	Use feeConfig consistently
TD-10	Slippage 0.5%	pages/Swap.tsx	Low	Hardcoded tolerance	Make configurable
TD-11	2FA secret `JBSWY3DPE`	pages/TwoFactor.tsx	High	Static secret	Per-user TOTP secret
TD-12	Crypto addresses	pages/Receive.tsx	High	Static deposit addresses	Per-user address generation
TD-13	NGN deposit bank	pages/Receive.tsx	Medium	Hardcoded Wema Bank code	Configurable admin setting
TD-14	Mock user data	contexts/AuthContext.tsx	Medium	MOCK_USER hardcoded	Remove in production
TD-15	Admin user	Admin pages	Medium	Hardcoded "Jonathan Tur" name	Real admin auth
TD-16	Withdrawal fees	pages/Withdraw.tsx	Medium	NGN=100, Crypto=\$0.50	Hardcode feeConfig
TD-17	Referral reward 5000	lib/referral.ts	Low	Hardcoded reward amount	Already in config, but also
TD-18	Promo banners	pages/Dashboard.tsx	Low	3 hardcoded banners	CMS-driven or admin-configurable

3. Repeated Logic

ID	Pattern	Locations	Description	Fix
TD-20	Number formatting	Multiple pages	Intl.NumberFormat repeated	Create formatCurrency() util
TD-21	localStorage key prefixes	All stores	cex_ prefix repeated manually	Create createStore() factory
TD-22	Coin icon URLs	CoinIcon.tsx, `Receive.tsx	External URL construction duplicated	Centralize in coin config

4. State Management Problems

ID	Issue	Severity	Description	Fix
TD-24	No state synchronization	Medium	Multiple tabs would have sta	Use storage event listener
TD-25	No state validation	Medium	localStorage data can be tam	Add schema validation on loa
TD-26	TanStack Query unused	Low	QueryClient created but no h	Either use it or remove it

5. Scalability Risks

ID	Risk	Severity	Description	Mitigation
TD-28	No pagination	High	Transaction list renders all	Implement virtualization or
TD-29	No lazy loading	Medium	All routes loaded upfront	Implement React.lazy() for r
TD-30	External image dependencies	Medium	cryptologos.cc and worldvect	Self-host or cache images
TD-31	No service worker	Low	No offline caching or PWA su	Add service worker for cachi

6. Security Risks

ID	Risk	Severity	Description	Mitigation
TD-33	No CSRF protection	Critical	No token-based auth	Implement JWT with CSRF toke
TD-34	No input sanitization	High	User inputs not sanitized ag	Sanitize all user inputs
TD-35	No rate limiting	High	No brute-force protection on	Implement rate limiting on a
TD-36	Hardcoded PIN	Critical	1234 is visible in source	Remove and implement proper
TD-37	No HTTPS enforcement	Medium	No mechanism to enforce secu	Enforce HTTPS in production
TD-38	No Content Security Policy	Medium	No CSP headers	Add CSP headers
TD-39	Admin route exposure	Critical	Anyone can access /admin/*	Add role-based route guards

7. Code Quality Issues

ID	Issue	Location	Severity	Description	Fix
TD-41	Missing loading states	Most pages	Medium	Only Dashboard and Hist	Add loading states to all da
TD-42	No TypeScript strict mode	tsconfig.json	Low	strict may not be fully en	Enable full strict mode
TD-43	Dead dependencies	package.json	Low	next-themes, cmdk, `vaul	Remove unused dependencies
TD-44	No CI/CD pipeline	N/A	Medium	No automated testing or dep	Set up GitHub Actions or sim
TD-45	Minimal test coverage	src/test/	Medium	Only `expect(true).toBe(tr	Add meaningful unit and inte

8. Actionable Fix Priority

Priority	Items	Effort	Impact
P1 — Sprint 1	TD-1, TD-2, TD-4, TD-7, TD-8	1-2 weeks	Foundation for production re
P2 — Sprint 2	TD-3, TD-19, TD-20, TD-21, T	1-2 weeks	Code quality and scalability
P3 — Sprint 3	TD-23, TD-24, TD-25, TD-29,	1 week	UX and performance improve
P4 — Ongoing	TD-31, TD-37, TD-38, TD-42,	Ongoing	Long-term quality and securi

Celler — Scope Control

Date: April 3, 2026

In Scope (Current Prototype)

Module	Features Included
Wallet & Dashboard	Portfolio balance (USD/NGN),
Crypto Swap	Coin selection, rate calcula
Withdrawal	Amount entry (custom numpad)
Receive	QR code display, address cop
Transaction History	Transaction list, type/statu
Gift Card Trading	Brand marketplace, sell flow
Referral Program	Referral code/link, earnings
KYC Verification	3-tier display, limits, subm
Profile & Security	Edit profile, password chang
Notifications	User notification list, admi
Admin Dashboard	Overview stats, customer man

Out of Scope (Current Prototype)

Category	Items
Real Auth	JWT, OAuth, email verificati
Blockchain	Wallet address generation, r
Payments	Payment gateway (Paystack/FI
Communications	Email service, SMS gateway,
File Storage	Document upload, image stora
Security	Rate limiting, CSRF protecti
Advanced Trading	Limit orders, stop-loss, DCA
Mobile App	Native iOS/Android app (Reac
Analytics	User behavior tracking, funn
Compliance	AML reporting, regulatory co
Internationalization	Multi-language support, mult
Business Features	White-label solution, B2B AP

Scope Boundary Rules

Rule	Description
Mock Data Only	All data operations use loca
Nigerian Market Only	NGN is the only fiat currenc
Prototype Purpose	This is a design/UX validati
No Real Money	No real cryptocurrency or fi
Single Admin Role	Only one admin role exists.

Scope Change Process

Any feature request that falls into "Out of Scope" must:

1. Be documented with business justification
2. Be assessed for effort and impact
3. Be approved by product owner
4. Be added to the roadmap (not the current sprint)
5. Have clear acceptance criteria defined

Celler — Product Recommendations

Date: April 3, 2026

1. UX Improvements

ID	Recommendation	Impact	Effort	Description
UX-R2	Fix swap fee discrepancy	High	Low	Swap.tsx shows 0.2% but conf
UX-R3	Implement error boundaries	High	Medium	Wrap app in React ErrorBound
UX-R4	Add pull-to-refresh on lists	Medium	Medium	Dashboard, History, Gift Car
UX-R5	Improve empty states	Medium	Low	Add helpful CTAs when no tra
UX-R6	Add confirmation dialogs for	High	Low	Logout, bank account removal
UX-R7	Implement toast notification	Medium	Medium	Currently only admin pages u
UX-R8	Add haptic feedback indicato	Low	Low	Visual feedback for button p

2. Feature Enhancements

ID	Recommendation	Impact	Effort	Description
FE-R2	Price alerts	Medium	Medium	Let users set alerts for coi
FE-R3	Recurring buys (DCA)	High	High	Dollar-cost averaging for au
FE-R4	Watchlist	Medium	Low	Let users track coins withou
FE-R5	Transaction categories/tags	Low	Medium	Let users categorize transac
FE-R6	Biometric login	Medium	Medium	FaceID/TouchID for quick app
FE-R7	Dark mode auto-schedule	Low	Low	Auto-switch theme based on t
FE-R8	Multi-language support	Medium	High	Yoruba, Hausa, Igbo alongsid

3. Monetization Opportunities

ID	Recommendation	Revenue Potential	Effort	Description
MON-R2	Premium subscription	High	High	Monthly fee for zero-fee tra
MON-R3	Gift card margin optimizatio	Medium	Low	Dynamic rates based on deman
MON-R4	Referral bonus for admin	Medium	Low	Admin-configurable referral
MON-R5	API access for businesses	High	High	B2B API for merchants to acc
MON-R6	Staking yield spread	Medium	High	Offer staking with a small p
MON-R7	Cross-sell insurance	Low	High	Partner with crypto insuranc

4. Performance Improvements

ID	Recommendation	Impact	Effort	Description
PER-R2	Image optimization	Medium	Medium	Self-host coin logos, use We
PER-R3	Virtualized transaction list	High	Medium	react-window for transaction
PER-R4	Service worker caching	Medium	High	Cache static assets and API
PER-R5	Debounce search inputs	Low	Low	Prevent excessive re-renders
PER-R6	Memoize expensive computati	Medium	Low	React.memo for coin list ite

5. Trust & Safety Improvements

ID	Recommendation	Impact	Effort	Description
TS-R2	Implement proper admin auth	Critical	High	Separate admin login with ro
TS-R3	Add transaction confirmation	High	Low	Show exact amounts, fees, an
TS-R4	Implement withdrawal cooldow	Medium	Medium	24-hour delay for new benefi
TS-R5	Add login notifications	Medium	Medium	Email/SMS alert on new devic
TS-R6	Session management page	Medium	Medium	Show active sessions, allow
TS-R7	Transaction velocity limits	High	Medium	Flag unusually frequent tran
TS-R8	Gift card duplicate detectio	High	Medium	Prevent same card code from
TS-R9	KYC document encryption	High	High	Encrypt uploaded documents a
TS-R10	Audit log for all admin acti	High	Medium	Track who did what and when

6. Quick Wins (High Impact, Low Effort)

#	Recommendation	Time
2	Remove hardcoded PIN (TD-6)	1 hour
3	Update index.html title and	15 min
4	Remove dead dependencies (TD	30 min
5	Add error boundaries (TD-4)	2 hours
6	Add loading skeletons to all	4 hours
7	Create useCurrency() hook (T	1 hour
8	Add confirmation dialogs (UX	2 hours
9	Fix 404 on Buy/Sell routes (	1 hour
10	Add admin route guards (TD-2	2 hours

Celler — One-Year Roadmap

Date: April 3, 2026

Current State: Frontend Prototype (v0.1.0)

Q1 — Stabilization & Core Foundation (Months 1-3)

Goals

- Build production-ready backend infrastructure
- Replace all mocked functionality with real services
- Establish security foundation

Features

Feature	Priority	Description
Database	P0	PostgreSQL with proper schem
Real Authentication	P0	JWT-based auth with Supabase
Admin Auth & RBAC	P0	Separate admin login with ro
Password Security	P0	bcrypt hashing, forgot passw
Session Management	P0	JWT with refresh tokens, dev
Buy Flow	P1	Payment gateway integration
Sell Flow	P1	Off-ramp integration for cry
Real Price Feeds	P1	CoinGecko or CoinMarketCap A
Email/SMS Service	P1	OTP delivery, transaction no

Technical Priorities

- Set up CI/CD pipeline (GitHub Actions)
- Implement comprehensive test suite (unit + integration)
- Add error boundaries and monitoring (Sentry)
- Establish API documentation (OpenAPI/Swagger)
- Set up staging and production environments
- Implement rate limiting and CORS policies

Success Criteria

- All critical gaps (G-1, G-2, G-3) resolved
- 80%+ test coverage
- Zero critical security vulnerabilities
- Successful end-to-end auth flow

Q2 — Core Feature Expansion (Months 4-6)

Goals

- Complete all core trading features
- Implement real blockchain integration
- Launch KYC verification system

Features

Feature	Priority	Description
NGN Payment Processing	P0	Real bank transfers via paym
KYC Verification	P0	BVN check, ID verification,
Address Validation	P1	Crypto address format and ch
Withdrawal Limits	P1	KYC-tier-based limit enforce
Transaction Pagination	P1	Virtualized list or paginati
Portfolio Chart	P2	Historical balance visualiza
Audit Trail	P1	Admin action logging
Multi-role Admin	P1	Super Admin, Support Agent,
File Storage	P1	S3/GCS for KYC documents and

Technical Priorities

- Implement WebSocket for real-time price updates
- Add service worker for offline caching
- Set up monitoring and alerting (Datadog)
- Implement database backups and disaster recovery
- Performance optimization (lazy loading, code splitting)
- API versioning strategy

Success Criteria

- All high-priority gaps resolved
- Real blockchain transactions working
- KYC processing live
- Admin dashboard fully functional with real data
- < 2s page load times across all pages

Q3 — Growth & Monetization (Months 7-9)

Goals

- Launch referral program with real tracking
- Implement fee collection and revenue tracking
- Expand gift card offerings

Features

Feature	Priority	Description
Fee Collection	P0	Real fee deduction and reven
Gift Card Fraud Detection	P1	Duplicate detection, image a
Push Notifications	P1	Real push notification deliv
Bill Payments	P2	Airtime, data, electricity b
Transaction Export	P2	CSV/PDF export for users and
Admin Reports	P1	Revenue, volume, and user gr
Live Chat Support	P2	Integrated support chat (Int
Multi-currency Support	P2	Additional fiat currencies b
Staking/Earn	P3	Crypto staking features

Technical Priorities

- Implement analytics pipeline (Mixpanel/Amplitude)
- A/B testing infrastructure
- Load testing and auto-scaling setup
- API rate limiting per user tier
- Database optimization and indexing
- Implement feature flags (LaunchDarkly)

Success Criteria

- Referral program driving 20%+ of new signups
- Revenue tracking accurate to the penny
- Gift card fraud rate < 2%
- Push notification delivery > 95%
- 10,000+ active users

Q4 — Scale & Optimization (Months 10-12)

Goals

- Optimize for scale and performance

- Advanced features and integrations
- Prepare for regional expansion

Features

Feature	Priority	Description
API for Partners	P2	Public API for third-party i
Mobile App (React Native)	P2	Native iOS/Android app
Advanced Trading	P3	Limit orders, stop-loss, DCA
Loyalty Program	P2	Tiered rewards based on trad
Tax Reporting	P3	Transaction history for tax
Multi-language Support	P2	Localization for West Africa
Compliance Reporting	P1	AML/KYC reporting for regula
White-label Solution	P3	Platform licensing for other

Technical Priorities

- Microservices migration (if monolith becomes bottleneck)
- CDN implementation for global performance
- Database read replicas for scaling
- Implement circuit breakers and graceful degradation
- Security audit and penetration testing
- SOC 2 / ISO 27001 compliance preparation
- Disaster recovery testing

Success Criteria

- 100,000+ registered users
- 99.9% uptime
- < 1s page load times globally
- Zero security incidents
- Ready for Series A fundraising

Roadmap Summary

Quarter	Theme	Key Deliverables	Risk Level
Q2	Core Features	Blockchain, KYC, Payments, A	High
Q3	Growth	Referrals, Monetization, Not	Medium
Q4	Scale	Analytics, Mobile, Complianc	Medium

Dependency Chain

- Q1 (Backend + Auth)
 - Q2 (Blockchain + KYC + Payments)
 - Q3 (Referrals + Monetization + Analytics)
 - Q4 (Scale + Mobile + Compliance)

Resource Requirements

Quarter	Engineering	Design	QA	DevOps
Q2	5-7 engineers	1 designer	2 QA	1 DevOps
Q3	4-5 engineers	1 designer	1 QA	1 DevOps
Q4	5-6 engineers	2 designers	2 QA	2 DevOps